

Executive Summary

Course: ECON 8310 | Business Forecasting

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1. The Goal: Why Track Baseballs?

Modern baseball is all about the numbers. For a local coach, being able to track how a ball moves, its speed, its path, and where it ends up, is huge for helping players get better. The problem is that watching hours of video by hand is exhausting and easy to mess up.

Our team's job was to build a computer brain that could watch these videos for the coach and automatically draw a tight box around the ball in every single frame. By turning messy video into clear data points, we give coaches a way to look at launch angles and pitch consistency without having to hit "pause" and "play" a thousand times.

2. Data Diagnosis: What Makes a Video "Work"?

A big part of our project was figuring out what actually makes a video easy for a computer to read. We call this data diagnosis.

The CVAT Headache: Dealing with Background Clutter

To teach our model what a baseball looks like, we used a tool called CVAT to manually box the balls in our training videos. One of the biggest hurdles we ran into was when the camera was positioned behind the batter. In those shots, there might be 30 or 40 other baseballs just sitting on the ground in the background.

The real technical friction happened when the batter would swing, and his arm or the bat would block those background balls for a split second. Our team had a choice: do we keep the box there while it's covered? Do we delete it and start over when it reappears? Or do we ignore the background balls entirely? Because the whole class couldn't really agree on a single rule, our team decided to focus strictly on the active ball in flight. By ignoring the stationary noise on the ground, we made sure our model didn't get distracted by clutter and stayed focused on the ball that actually matters to the coach.

Our "Wish List" for Better Data

To make our tracking more solid, we requested specific types of videos:

- **High-Contrast Backgrounds:** White balls disappear against light colors. We found that dark screens or green turf make the ball pop, helping the model see it much more clearly.
- **Different Angles:** We didn't want the model to only recognize a ball from one spot. By asking for catcher's view and side-profile shots, we taught the model to understand the ball's size as it moves closer or further away.
- **High Frame Rate (FPS):** Fast balls look like blurry white streaks in normal video. High FPS keeps the ball looking sharp, which is the only way to get those tight bounding boxes the rubric asks for.

3. Our Strategy: How the Model Thinks

We used a mix of old-school motion detection and new-school "Deep Learning."

Frame Differencing: Cutting Out the Noise

To help the model in messy environments (like inside a hitting cage with lots of nets), we used Frame Differencing. Think of this as the model subtracting everything that stays still. If the net and the ground don't move, the model ignores them and only looks at the pixels that are actually moving. This makes the search area way smaller and the tracking much faster.

Transfer Learning: Giving the Model a Head Start

Instead of starting from zero, we used Transfer Learning. We started with a model called ResNet-18, which was already pre-trained on millions of images to recognize basic things like shapes and edges. We then fine-tuned it specifically on our baseball data. A major win for our team was successfully re-wiring the model to output four specific numbers (x, y, width, and height) so it could draw a physical box rather than just guessing what the object was.

The Tightness Test: IoU

The most important metric for us was Intersection over Union (IoU). For the coach, this is basically the accuracy grade. It measures how well our predicted box overlaps with where the ball actually is. A high IoU means our boxes are tight and reliable for professional use.

4. Putting it to Work for the Coach

At the end of the day, a coach doesn't want to see (x, y) coordinates; they want to see how their player is doing. Our analysis turns those numbers into real-world insights:

- **Pitch Break:** By tracking the (x, y) path, a coach can see exactly how much a curveball drops. If a pitcher is tired, that 6-inch drop might turn into a 3-inch drop, and our data provides the proof.
- **Strike Zone Maps:** We can layer the coordinates from multiple pitches to create a heat map, showing if a pitcher is consistently hitting their spots or if they are drifting as the game goes on.
- **Release Point:** We can find the exact coordinate where the ball first appears, helping coaches make sure a pitcher isn't tipping their pitches by changing where they let go of the ball.

5. Final Conclusion

By bridging the gap between raw, messy video and automated tracking, we've built a tool that lets coaches spend less time at a computer and more time on the field. Everything we built is designed for replication, with our Google Colab link, the instructor can run our entire workflow from start to finish and see the results immediately without any errors.